Enver Gökay ÇAY

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Education

Karadeniz Technical University, Electrical and Electronics Engineering (100% English) 2020 – 2025

Experience

Hardware Design Engineer, Casfo GmbH – Stuttgart (Remote) May 2026 – Present

- Development of **system power architecture (Power Tree)** and **battery management systems (BMS)**; design of power management, transient protection and **DC-DC converter circuits** for 12V/24V vehicle power supply lines.
- Hardware and communication interface design for **KNX-based** smart building automation solutions.
- Integration of **NFC**, **GNSS**, and **LTE modules** and implementation of **RF matching network** and **antenna designs**.
- Design of secure hardware architectures including cryptographic components and physical security mechanisms.
- Hardware-level design and execution of validation processes for **SPI**, **I2C**, **UART**, and **CAN** communication interfaces.
- Execution of PCB layout considering **signal integrity**, **EMI/EMC** and **design for manufacturability (DFM)** criteria.
- Providing technical support for component selection, prototyping, transition to production, and validation processes.

Projects

3-Phase BLDC Motor Driver Board

- Conducted end-to-end hardware design of an **STM32G474-based 3-phase BLDC motor driver** board, including **DRV8353 gate driver** integration, power stage design for **SVPWM/FOC motor** control architecture, current-voltage measurement circuits, **EMI filtering**, and development of a **6-layer high-current PCB**.

  Project Link: https://github.com/envergokaycay/3-phase-bldc-motor-driver

Wireless Charging System Design for Electric Vehicles (Graduation Project)

- Developed the power electronics architecture, driver circuits, and PCB design of a **DC-AC inverter** circuit using **PWM switching techniques** with **H-Bridge topology**.

  Project Link: https://github.com/envergokaycay/h-bridge-dc-ac-inverter

STM32F417 Demo Board

- Developed a multi-purpose embedded system platform including USB, SWD programming/debugging infrastructure, power management, communication interfaces, and peripheral integrations by designing a development board.

  Project Link: https://github.com/envergokaycay/STM32-Demo-Board

Technical Training

High-Frequency Electromagnetic Analysis Training | Numesys

- Performed design and optimization of a **2.4 GHz PIFA antenna (4.01 dBi gain, -34.2 dB S11)** for smart systems and communication using **Ansys HFSS**.
- Conducted **3D electromagnetic analyses** and **EMI simulations** of high-speed circuit structures.

  Project Link: https://github.com/envergokaycay/2.4GHz-PIFA-Antenna

AI-Based PCB EMI Defect Detection System | Samsung Innovation Campus

- Developed an AI model that detects PCB EMI/SI design defects with **93% accuracy (YOLOv8, Python)**.

  Project Link: https://github.com/envergokaycay/ai-pcb-emi-signal-integrity

Technical Skills

- **PCB Design and Simulation:** Altium Designer, KiCad, Multi-layer Design, High-Speed Routing, Signal Integrity, EMI/EMC, DFM, LTspice
- **Power Electronics:** Buck/Boost/Buck-Boost Converters, BLDC Motor Drives, System Power Architecture (Power Tree), Battery Management Systems (BMS)
- **RF and Microwave:** Ansys HFSS, NFC, GNSS, LTE, Antenna Integration, RF Impedance Matching
- **Embedded Systems:** STM32, SPI, I2C, UART, CAN Bus, KNX Protocol
- **Languages**: Turkish (Native), English (Advanced)